

Analysis theory and engineering applications of steel–concrete hybrid tower structures for large wind turbines

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ABSTRACT

Developing wind power is crucial for achieving China's dual-carbon goals. In recent years, the wind power industry has faced trends of increasing turbine capacity, significantly taller hub heights, and longer blade lengths. Traditional steel towers, owing to their high costs and frequent accidents, are becoming less suitable for meeting these demands. As a result, steel–concrete hybrid high-tower structures (referred to as “hybrid high-tower structures”) have emerged as the primary support for large-scale wind turbines. This paper presents the latest research developments in steel-prestressed concrete hybrid towers and prestressed concrete-filled steel tubular (CFST) lattice towers, which play pivotal roles in advancing hybrid high-tower technology.

1 Introduction

Wind energy has emerged as one of China's most vital sources of clean and renewable energy. Expanding wind power generation represents a critical step for advancing China's energy production technologies and driving industrial innovation. This aligns with China's strategic goal of developing a resource-efficient and environmentally friendly society. Currently, China has the world's largest installed capacity for onshore wind power [1]. As the scale of wind power installations continues to expand, subsidy policies for new wind power projects have been phased out. In line with the “grid parity” requirements, the wind power industry in China now faces increasing demands for higher turbine capacity, greater hub heights, and larger blade diameters.

Traditional steel towers, typically fabricated by rolling and welding steel plates, offer high levels of

standardization, a mature industrial supply chain, and rapid installation. As a result, they are commonly used as support structures for wind turbines with hub heights less than 100.0 m. However, limitations in transportation and production costs constrain the load-bearing capabilities of traditional steel towers. Consequently, employing steel towers for large-capacity wind turbines with high hubs and long blades presents several challenges [2, 3]: low natural frequency, which may coincide with the turbine's operating frequency, increasing resonance risk; limited lateral stiffness, causing excessive top displacement and prominent fatigue issues; restricted load-bearing capacity, where increased turbine loads may induce local instability. Recently, the number of steel tower collapses has increased (Fig. 1), resulting in significant economic losses and hindering the development of large wind turbines.

Hybrid high-tower structures refer to designs where

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Fig. 1 Sites of the pure steel tower collapse incident.

steel towers are used in upper regions with lower load demands, whereas lattice towers [4, 5] or prestressed concrete towers [6] are employed in lower regions with higher load demands, thereby overcoming the technical limitations of traditional steel towers. These hybrid towers provide solutions to issues such as local buckling, low lateral stiffness, and high steel costs, and they have recently been applied at the industrial scale in China, as exemplified by projects in Shandong and Henan Provinces. However, as hybrid tower technology advances, the engineering design process faces challenges, including the absence of a unique innovation framework, a lack of foundational theoretical models, and outdated analytical techniques. Therefore, this paper presents novel research findings on two types of hybrid tall towers, focusing on their structural configurations and load-bearing characteristics.

2 Steel-prestressed concrete hybrid towers

Steel-prestressed concrete hybrid towers, with hub heights ranging from 120.0 to 160.0 m, offer considerable benefits, including high rigidity, excellent stability, and low maintenance costs. These features markedly increase power generation efficiency in wind farms located in low-wind-speed areas, aligning well with China's onshore wind energy development needs and the industry's shift toward grid parity. The structural design is as follows (Fig. 2): The lower portion of the tower, which bears greater stress, is constructed using a concrete tower structure. Since the base diameter exceeds 8.0 m, surpassing the onshore transport limit of 4.5 m for component size, the structure employs a factory-prefabricated, onsite assembly method, effectively addressing transportation constraints. Vertical full-length prestressing is applied to ensure that the entire cross-section of the concrete structure remains in compression. The upper section, which experiences lower stress, utilizes a steel tower structure, which is connected to the concrete base via a transition structure. Many wind farms have already implemented this type of steel-prestressed concrete hybrid tower (Fig. 3).

Unlike traditional civil buildings, the primary consideration in tower design is to avoid resonance with the operational frequencies of the wind turbine units, making it essential to establish an efficient method for calculating

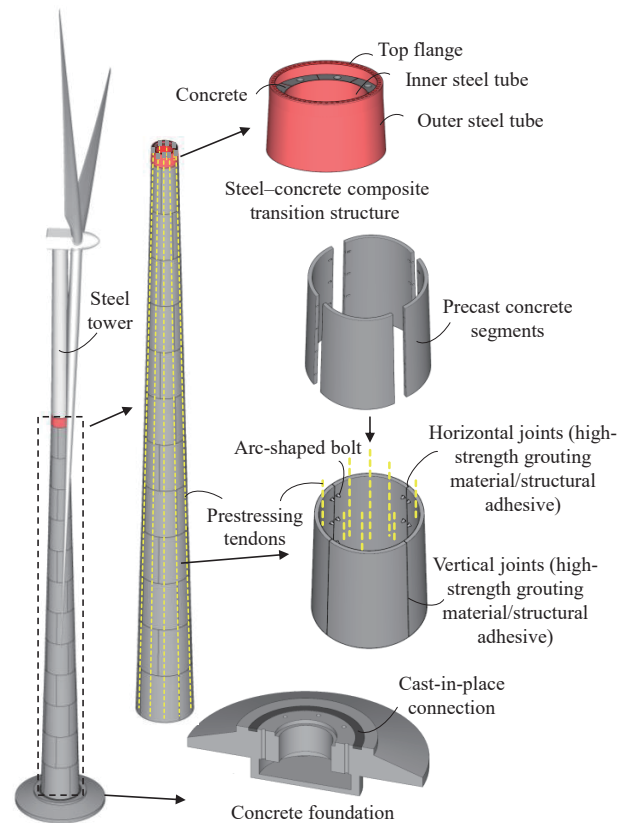


Fig. 2 Steel-prestressed concrete hybrid towers.

the fundamental frequency of hybrid high tower structures. Additionally, wind power structures, characterized by their tall towers and large-blade sweep areas, experience complex coupled vibrations due to varying loads throughout their service life. As the turbine size increases, additional bending moments caused by the second-order gravity effect can amplify vibrations at the tower's top. Currently, research on the dynamic responses of wind power structures predominantly addresses pure steel towers. However, with the transition to hybrid high-tower structures, significant changes occur in the overall structural assembly and vibration characteristics. Therefore, investigating the dynamic response and implementing damage identification methods for hybrid high tower structures is of particular importance.

An efficient computational method for determining the natural frequency of hybrid high-tower structures is proposed on the basis of theoretical derivation, offering a practical approach to control the natural vibration fre-

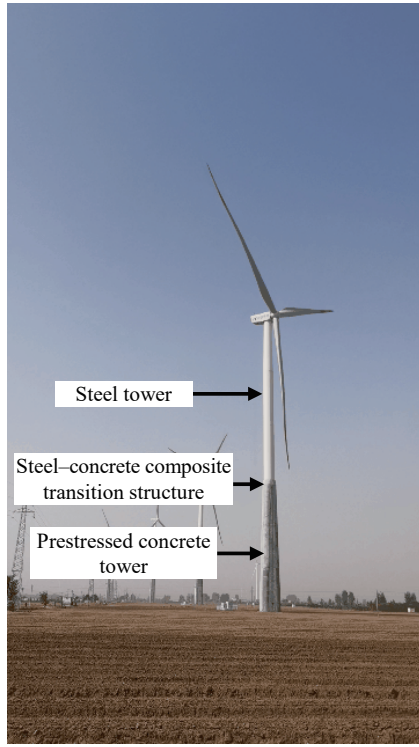


Fig. 3 Application of steel–prestressed concrete hybrid towers.

quency during preliminary design phases and scheme adjustments [7]. The theoretical derivation process includes deriving expressions for the kinetic and potential energy of the structural system while considering factors such as the mass and rotational inertia of the wind turbine, elastic foundation stiffness, and impact of prestress in the concrete tower segment on the mechanical energy of the structure (Fig. 4). In Fig. 4, m_{wt} and J_{wt} denote the mass and rotary inertia of the wind turbine, respectively; d is distance from the centroid of the wind turbine to the top end of the tower; P is prestressed force; $u(x, t)$ is transverse displacement of the steel–prestressed concrete hybrid towers; t denotes the time; θ denotes the angle between AB and $A'B'$; H is the height of the steel–prestressed concrete hybrid towers; u and x denote

the transverse and longitudinal coordinate, respectively. Using the Rayleigh–Ritz method, a frequency equation is derived, and n orthogonal polynomial groups are generated over the height interval via the Gram–Schmidt transformation. These polynomials serve as approximate shape functions for the steel–concrete hybrid tower. The Gram–Schmidt orthogonal polynomials are subsequently substituted into the frequency equation to solve for the fundamental frequency.

To investigate the convergence of the proposed computational method, polynomial functions with different structural forms were selected as the initial term $\phi_0(x)$ of the Gram–Schmidt orthogonal polynomial set to construct the mode shape function $y(x)$. The convergence analysis of the fundamental frequency is shown in Fig. 5. As the number of orthogonal polynomials in the mode shape function increases, the natural frequency decreases and approaches a convergent value. Discrepancies between the selected mode shape function $y(x)$ and the actual mode shape function introduce additional constraints on the structure, resulting in an overestimation of the natural frequency. As the number of orthogonal polynomials in the mode shape function increases, its shape gradually approximates the true mode shape function, causing the natural frequency from the theoretical solution to decrease and converge. Thus, the proposed solution method provides an upper bound of the exact solution.

A series of numerical examples involving hybrid tower structures with varying design parameters were utilized to validate the proposed fundamental frequency calculation method by comparing its results with those obtained from finite element analysis. The comparison reveals that the maximum relative error between the proposed calculation method and the numerical simulation results is only 0.143%, demonstrating the method's accuracy. Furthermore, the method was employed to investigate how the fundamental frequency of the hybrid tower structure responds to changes in key design parameters, as illustrated in Fig. 6. The three parameters that most

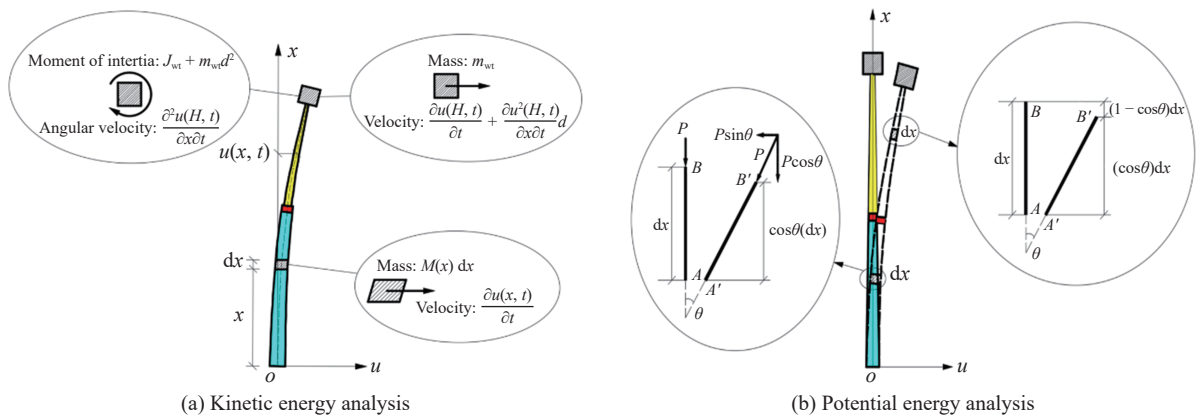


Fig. 4 Schematic diagram of the energy analysis for the hybrid tower.

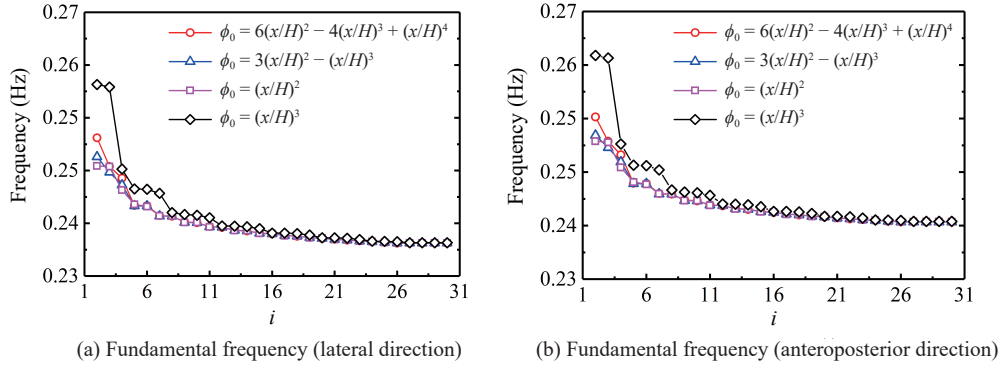


Fig. 5 Convergence analysis of the fundamental frequency calculation results (i is the number of polynomial terms).

significantly influence the fundamental frequency are l_c , t_{cb} , and t_{sb} , with the design parameters of the steel tower and transition segments exerting a markedly smaller influence than those of the concrete tower segment. As the overall height of the structure increases, the correlation between the fundamental frequency and the length ratio l_c becomes more pronounced. Specifically, as the total height decreases from 160.0 to 140.0, 120.0, and 100.0 m, the linear correlation coefficient for l_c decreases by 4.1%, 15.4%, and 103.7%, respectively. For a tower with a total height of 100.0 m, the primary influencing parameters for the structural fundamental frequency are t_{cb} and d_{cb} .

A method for inverse analysis of the dynamic response of wind turbine tower structures and the horizontal thrust exerted by the turbine, which is based on Kalman

filtering theory, was proposed [8]. This approach aims to estimate the dynamic response at various elevations of the tower and time series data of the horizontal thrust at the tower top using only the measured dynamic response data from a limited number of positions on the tower. Initially, the motion differential equations for the hybrid tower structure in physical coordinates are decoupled on the basis of the analysis of modal shapes and natural frequencies, after which the decoupled motion equations are transformed into state space. A coefficient matrix is subsequently constructed to modify the state equations and system noise, ensuring compliance with the fundamental assumption of independence between system noise and observation noise, as stipulated in Kalman filtering theory. Finally, Kalman filtering is applied to estimate the state variables, yielding modal

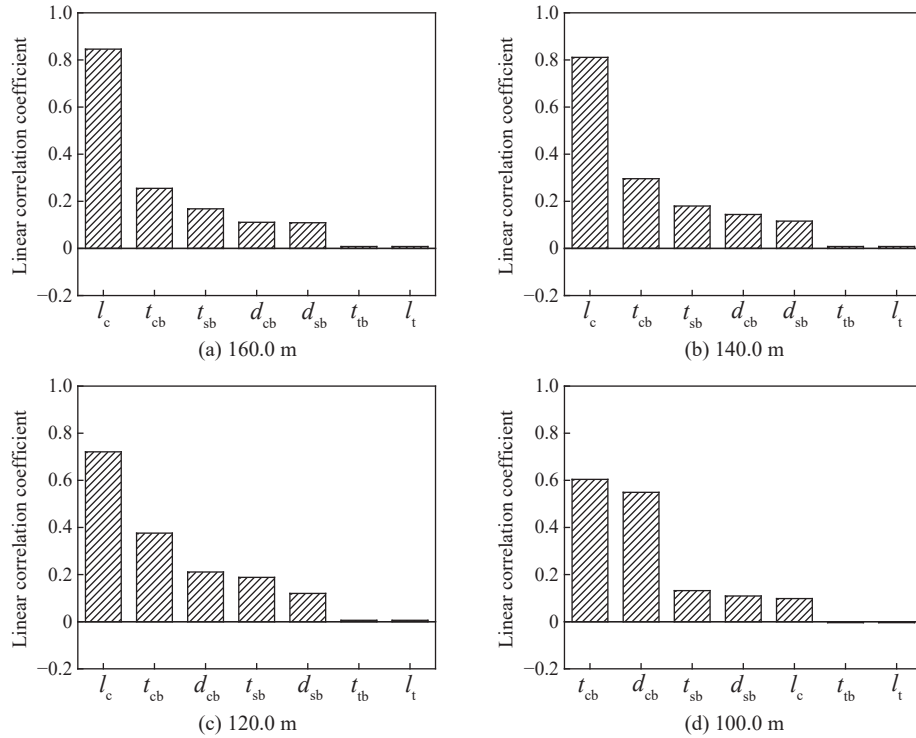


Fig. 6 Linear correlation coefficients between design variables and baseline rates. (l_c and l_t : length of the concrete portion and adapter portion; t_{cb} and t_{sb} : bottom thickness of the concrete portion and steel portion; d_{cb} and d_{sb} : bottom diameter of the concrete portion and steel portion; t_{tb} : bottom thickness of the adapter portion).

load and modal response estimates, and subsequently, predictions of the horizontal thrust of the turbine and the dynamic responses at locations beyond the measurement points on the hybrid tower.

In a numerical example of a hybrid tower, the dynamic response at a specific height of the tower is used as input for the inverse analysis. The resulting time history curve of the horizontal thrust exerted by the wind turbine on the hybrid tower and the root mean square (RMS) of the dynamic responses at various heights, obtained through the proposed inverse analysis method, are presented in Figs. 7 and 8, respectively. Importantly, the simulated values for the turbine's horizontal thrust and the dynamic responses at different heights were derived from numerical simulations via Bladed software and ABAQUS software. The labels $E_p = 1\%$ and $E_p = 5\%$ indicate the addition of white noise sequences with measurement errors at 1% and 5% noise levels in the input values for the inverse analysis.

A comparison between the inversion estimates and simulation values reveals that the proposed inversion analysis method accurately estimates the horizontal thrust and dynamic responses (displacement, velocity, and acceleration) of the wind turbine at unmeasured locations in hybrid tower structures. This method maintains a high inversion accuracy even at a noise level of 5%, offering a valuable reference for subsequent load calcu-

lations and structural design optimization of hybrid tower structures.

3 Prestressed concrete–filled steel tubular lattice towers

The lattice tower is designed to convert the overall bending mode of the tower structure into an axial force mode in the corner components, as shown in Figs. 9 and 10. To meet the strength and stability requirements, the hollow sections used in the design of the steel towers would require a tube wall thickness of over 25 mm. Achieving this thickness with the necessary precision would demand a specialized plate-bending machine, along with careful attention to the welding quality at the joints, thereby increasing the structure's cost. In recent years, concrete-filled steel tubular (CFST) members have been widely adopted in various structural applications, including transmission towers, building frameworks, offshore structures, and bridges. The popularity of these materials is due to their ease of manufacturing, high load-bearing capacity, cost efficiency, and excellent ductility [9–12], contributing to more economical structural design and construction.

The tensile strength of concrete is relatively low. Hence, the steel tube in CFST structures primarily provides tensile resistance, often necessitating a significant

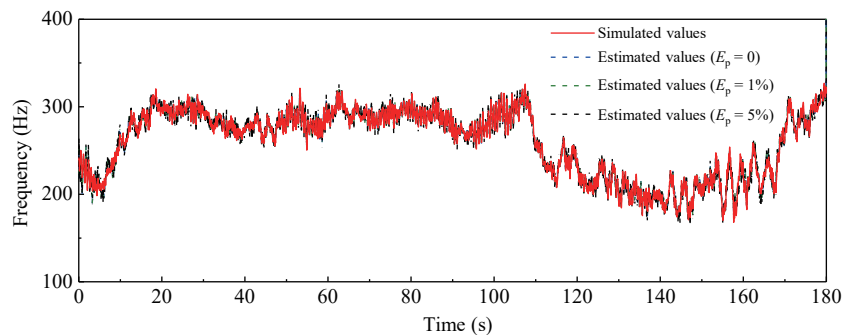


Fig. 7 Inversion analysis results of horizontal thrust on the wind turbine.

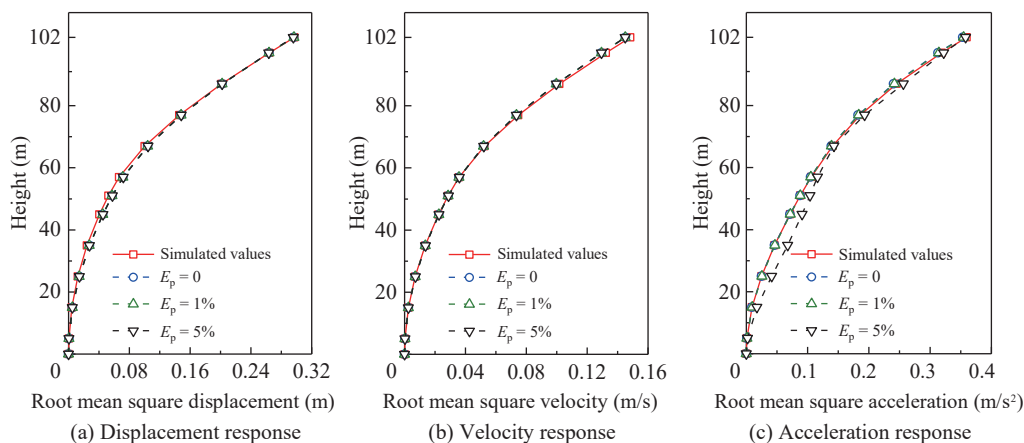


Fig. 8 Analysis results of the dynamic response inversion.

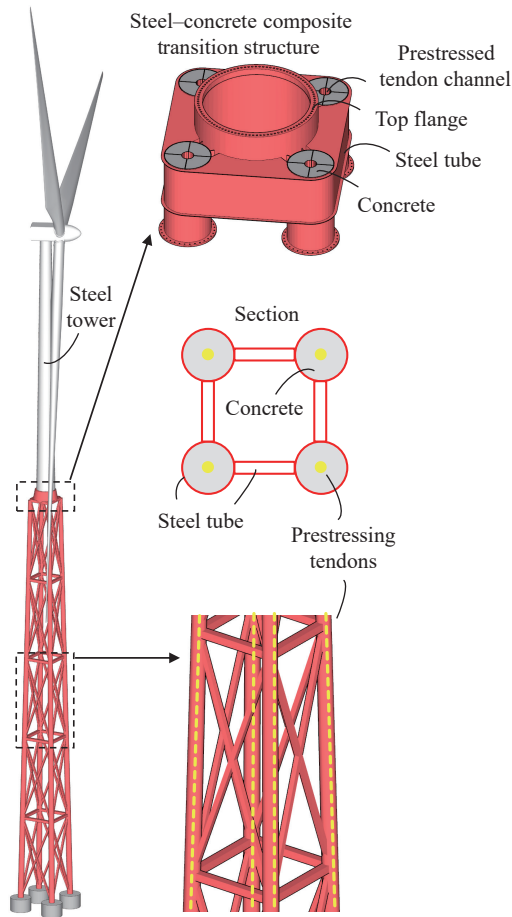


Fig. 9 Prestressed concrete-filled steel tubular lattice towers.

amount of steel. To address this, the concept of prestressed CFST lattice structures (Fig. 9) is proposed. By applying prestressing forces, the CFST columns are kept predominantly in compression during wind turbine operation. This approach leverages the high compressive load-bearing capacity of CFST columns while minimizing tensile forces on the members. Increasing the concrete cross-section rather than the steel tube wall thickness enhances the load-carrying capacity and reduces the required steel quantity, resulting in cost savings. The lower section of the tower structure is designed with a lattice configuration to withstand higher loads and is connected

to the upper steel tower through a transition piece (Fig. 10b).

3.1 Prestressed system design

In the lattice tower structure, the corner columns are constructed via concrete-filled double-skin steel tubular (CFDST) columns (Fig. 9), with the hollow section serving as a prestressing duct. Prestressed tendons are routed from the top of the transition piece down to the foundation, where they are tensioned at the top of the transition piece. This setup ensures a secure connection between the lattice section and the transition piece, facilitating the uniform transfer of prestress to the lattice columns. The magnitude of prestressing plays a central role in the structural design strategy, with its effects primarily reflected in three aspects: (1) allowing the components to primarily bear compressive forces, thus fully utilizing the high compressive load-carrying capacity of the CFST column; (2) preventing cracking in the internal concrete during normal operation of the wind turbine to preserve the load-bearing capacity of the structure; and (3) increasing the overall stiffness and stability of the structure.

Additionally, according to concrete fatigue design codes [13], the fatigue performance of concrete decreases under high compressive stress levels. Therefore, optimizing structural design principles is essential to minimize material costs. On the basis of the structural load characteristics and performance requirements, each corner column in this project is prestressed to ensure that the internal concrete remains below its tensile strength, preventing cracking during normal operation of the wind turbine.

3.2 Joint design

The diagonal and horizontal braces are connected to the corner columns via gusset plates (Fig. 11c), whereas adjacent columns are connected via flanges (Fig. 11e). The components are assembled with ring groove rivets, which provide excellent resistance to fatigue and loosening, ensuring maintenance-free operation of the struc-

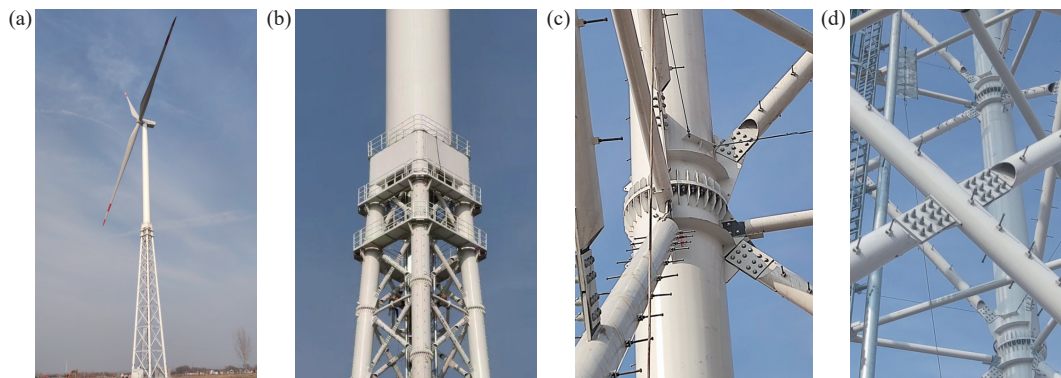


Fig. 10 Structural members and connections of the lattice wind turbine tower: (a) overall diagram; (b) transition piece; (c) joint region; and (d) diagonal braces.

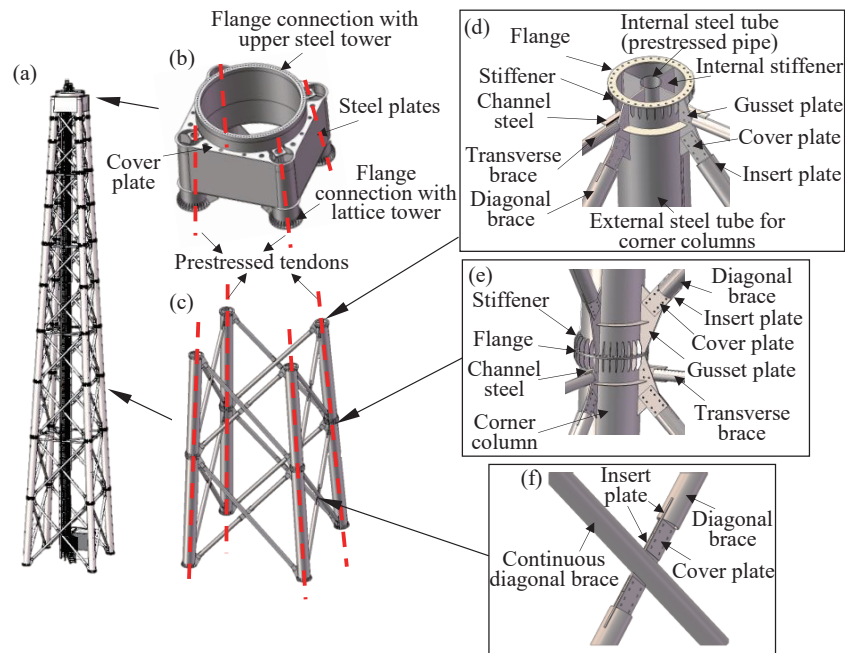


Fig. 11 Schematic diagram of the lattice part and transition piece: (a) lattice part; (b) transition piece; (c) section; (d) internal structure of joint region; (e) joint region; and (f) diagonal braces.

ture during its designed 20-year service life. Additionally, ring groove rivets allow for efficient assembly, with each rivet taking 5–8 s to install, significantly improving the overall assembly process.

At the ends of the corner columns, internal stiffeners are used to position the inner steel tubes and improve the load-bearing capacity of the joint region, as shown in Fig. 11d. The gusset plates at the ends of the angle columns extend into the concrete and are welded to the inner steel tubes, with the studs placed on either side of the gusset plate within the concrete. This construction ensures efficient load transmission, allowing part of the load to be directly transferred to the concrete and inner steel tubes through the gusset plate. This approach prevents the outer steel tubes from buckling under tension, improving both the fatigue performance and load-bearing capacity of the joint.

The dimensions and quantities of flange bolts should be determined to ensure that the flange surfaces of the wind turbine remained in constant compression during normal operation. It is essential to consider the additional bending moment induced by flange deformation under load when designing flange bolts without stiffeners [14].

3.3 Transition piece design

The transition piece, which serves as the connection structure between the steel tower and the lattice section, is crucial for the safe and reliable operation of the entire wind turbine. This allows for a reliable connection between the upper steel tower and the lower lattice section, enabling them to function cohesively. The most commonly used transition piece is the reinforced concrete

transition (RCT) piece, which is technologically mature and widely applied. However, RCT has several drawbacks: complex manufacturing and challenging assembly due to densely placed steel reinforcements; susceptibility to cracking, which can affect the service life of the tower; relatively low load-bearing capacity. Therefore, there is an urgent need to enhance the structural performance of the transition piece to address these limitations.

To improve the mechanical performance of RCT, a novel type of steel–concrete composite transition piece (SCCT) has been developed. The SCCT consists of steel components surrounding sandwiched concrete (Fig. 11b). In the design, the four corner columns of the lattice structure extend up to the top of the transition piece, whereas the steel tube of the upper steel tower extends downward to the bottom of the transition piece. The corner columns are positioned on the outside of the steel tube, with conventional stiffeners evenly distributed between them. Additionally, cover plates are installed on the outer side, top, and bottom of the transition piece, with additional reinforcing plates uniformly welded between the outer cover plate and steel tube. Finally, the cavities are filled with concrete to enhance the composite effect and overall structural integrity.

Importantly, under bending moments, the steel plates on the tension side of the transition piece are prone to detachment from the concrete, as well as cracking of the concrete. To prevent these issues, the studs are uniformly distributed on the inner side of the steel components within the transition segment. This further enhances the overall bond between the steel and the concrete. The upper end of SCCT is fastened to the steel tower via

flanges, whereas the lower end is connected to the lattice structure through flanges affixed to the four corner columns. The side width of SCCT is set at 5.0 m, taking into consideration the transportation width limitations on land.

The SCCT enhances the load-bearing capacity through the composite performance of concrete and steel. Furthermore, the use of steel tubes reduces the amount of steel reinforcement needed, eliminating the need for traditional formwork and steel reinforcement assembly processes associated with conventional RCT and thereby improving manufacturing efficiency. The SCCT also helps prevent concrete cracking, ultimately extending the service life of wind turbines.

Given the limited application of SCCT in wind turbine structures, research on this topic remains sparse. The structural configuration of SCCT used in this example resembles that of CFST, especially CFDST. Therefore, previous research on the compressive performance of CFSTs [15] and CFDSTs [16–21] can provide valuable information for investigating SCCT.

While previous research on CFST and CFDST structures provides valuable insights, it is essential to recognize that the geometric characteristics of these structures differ from those of SCCT. Therefore, to bridge the research gap in wind turbine structures and establish a theoretical foundation for using SCCT in wind turbine structures, both experimental and theoretical investigations have been conducted to evaluate the compressive performance of SCCT [6]. Using validated finite element models, a parameter analysis was performed to identify the key parameters influencing the load-bearing capacity of SCCT. This led to structural optimization aimed at reducing overall costs.

3.4 Structural investigations

3.4.1 CFDST columns

A study was performed on the mechanical performance of a CFDST column under different loading conditions. A total of 59 samples were tested under axial compression, bending, torsion, compression–torsion, and compression–bending–torsion loads [22–26].

To simulate complex loading conditions involving compression, bending, and torsion on the samples, an innovative compression–bending–torsion loading device was designed (Fig. 12). This device consists of components such as a steel triangular frame, jack, rigid girder, rigid rods, bidirectional hinge, actuator, and steel pedestal. It enables the simultaneous application of pure torsion, bending, compression–bending, and compression–bending–torsion loads.

The CFDST column samples displayed ductile failure characteristics, indicating strong load-bearing capacity and deformability. The typical failure modes are shown in Fig. 13. F represents the force, and the loading method is illustrated in Fig. 13. At the ultimate state, the interlayer concrete was crushed, and both the inner and outer steel tubes yielded. The failure mode involved outward bulging of the outer steel tube, inward buckling of the inner steel tube, and crushing of the interlayer concrete.

To enhance the mechanical performance of the CFDST columns, additional stiffeners were incorporated within the columns. A total of 7 CFDST columns with different stiffening types (studs and ribs) were tested under axial compression [27, 28], as shown in Fig. 14. During the experiments, two distinct failure modes were observed in the outer tubes: oblique buckling and outward circumferential buckling. Concurrently, localized crush-

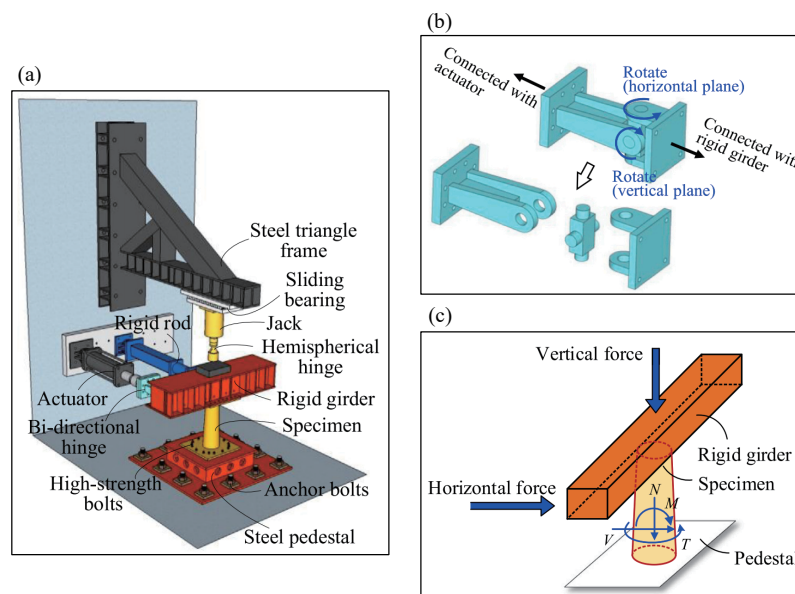


Fig. 12 Test setup of the samples under combined loads: (a) schematic view of the setup; (b) details of the bidirectional hinge; and (c) schematic view of the loading mode.

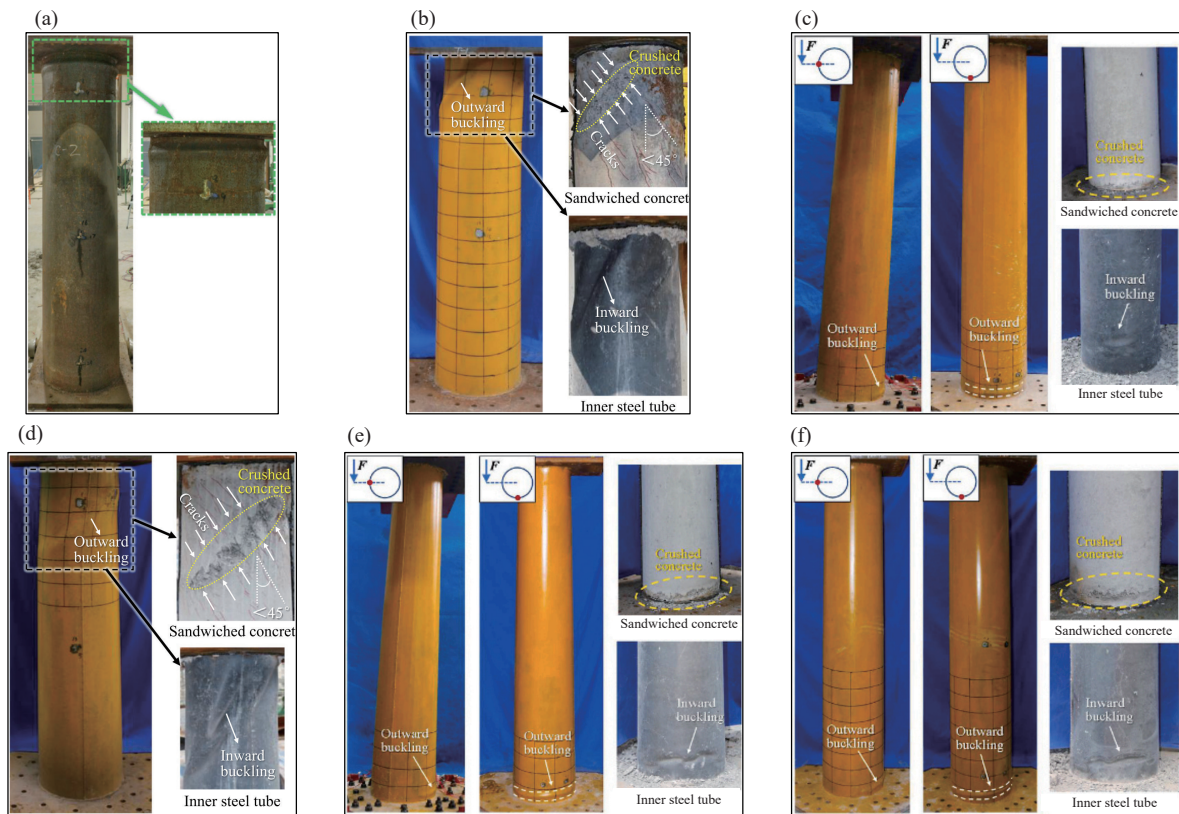


Fig. 13 Failure modes of the samples: (a) axial compression; (b) pure torsion; (c) bending; (d) compression–torsion; (e) bending–torsion; and (f) compression–bending–shear–torsion.

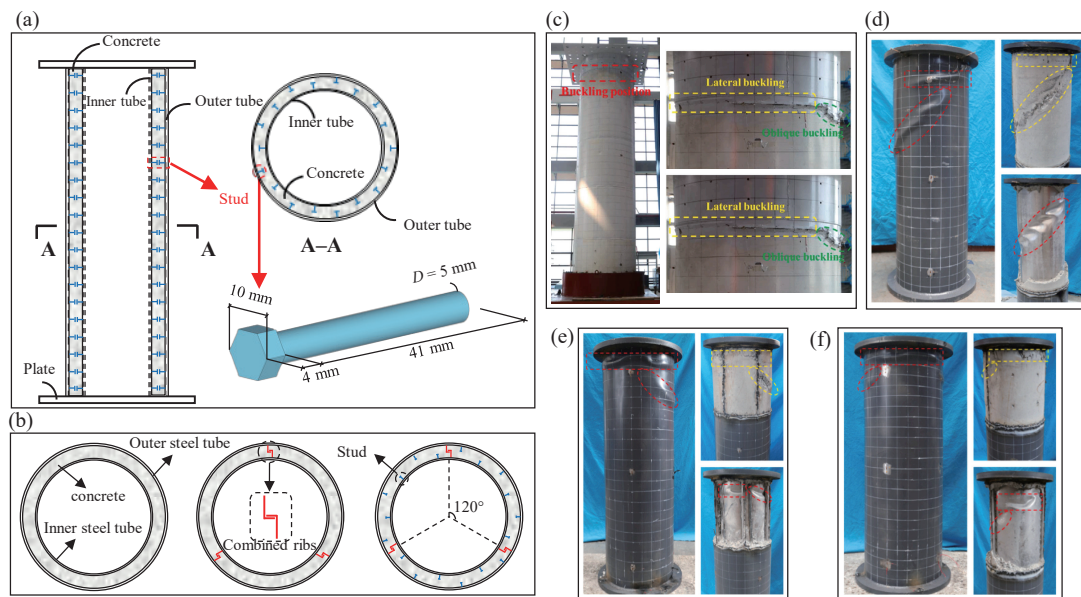


Fig. 14 Test samples of CFDST columns with different forms of stiffening: (a) studs (D represents the stub diameter); (b) unstiffened, combined ribs, and combined + studs; (c) failure mode of the studs; (d) failure mode of the unstiffened specimen; (e) failure mode of the combined ribs; and (f) failure mode of the combined ribs + studs.

ing of the interlayer concrete was evident, whereas the inner steel tubes exhibited inward buckling. The bearing capacity of the inner and outer tubes exhibited minimal variations with different stiffening configurations. Conversely, the bearing capacity of the interlayer concrete improved noticeably as the number of stiffened rib groups increased. Compared with those without stiffen-

ers, the load capacity of the samples with six sets of stiffened ribs was nearly 20% greater. Moreover, the performance of the samples containing additional studs improved even further.

3.4.2 Lattice part

The seismic performance of prestressed CFST latticed

towers was investigated through both experimental and theoretical research [5]. Two prestressed CFST latticed towers were constructed, and seismic tests were conducted under combined compression–bending–torsion cyclic loads, as shown in Fig. 15. The samples were fabricated at a scale of 1:10 to replicate the actual dimensions of the prestressed CSFT latticed wind turbine tower.

Under combined compression–bending–torsion cyclic loads, the diagonal braces of the prestressed CFST latticed tower experienced severe damage, irrespective of the presence of prestressing tendons. In the sample without prestressing tendons, the diagonal braces failed first, followed by damage to the CSFT corner columns. In contrast, the samples with prestressing tendons primarily experienced failure in the diagonal braces, with minimal visible damage to the CFST corner columns. Additionally, few cracks or signs of crushing were observed in the concrete core.

The application of prestressing tendons effectively suppressed concrete cracking. The hysteresis curves of the samples with prestressing tendons exhibited a plumper shape, indicating superior energy dissipation capacity, initial stiffness, and structural ductility. Thus, the prestressed CSFT latticed tower exhibits excellent seismic performance.

3.4.3 Joint

Static tensile, fatigue, and fatigue damage tests followed by static tensile tests were conducted on tube–gusset

plate joints [4, 29]. Three types of tube–gusset plate joints were designed: a tube–gusset plate joint with two ring stiffener plates (T-PRs), a CFST–gusset plate joint with no ring stiffener plate (CFST-P), and CFST-PRs. On the basis of experimental, numerical, and theoretical studies, the mechanical performance of these joints under static tensile, fatigue, and fatigue damage followed by static tensile loading was analyzed, as illustrated in Fig. 16.

Under tensile loading, specimen T-PR experienced local buckling of both the ring stiffener plate and the steel tube, resulting in ductile failure. However, specimen CFST-P exhibited shear failure of the steel tube wall, indicating brittle failure. Specimen CFST-PR demonstrated shear failure of the steel tube wall along with partial V-shaped buckling of the ring stiffener plate under loading. The results showed that the ring stiffener plate significantly enhanced the bearing capacity of the tube–gusset plate joint and ductility. Compared with the ring stiffener plate alone, the internal filling of the concrete was more effective in enhancing the bearing capacity. Specimen CFST-PR demonstrated the combined advantages of both the ring stiffener plate and the concrete, substantially improving the bearing capacity while ensuring the ductility of the joint. The experimental results indicate that, compared with those of specimens CFST-P and T-PR, the peak capacity of CFST-PR increased by 46.8% and 129.1%, respectively.

The results from stress concentration factor (SCF)

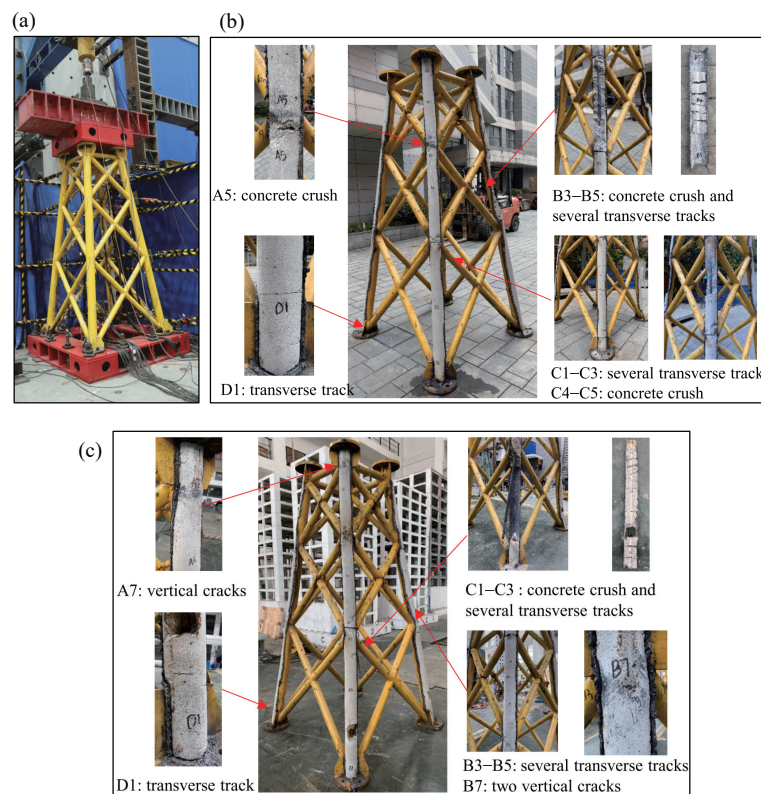


Fig. 15 Test samples of the prestressed CFST lattice tower: (a) test setup; (b) failure mode for the sample without prestressing tendons; and (c) failure mode for the sample with prestressing tendons.

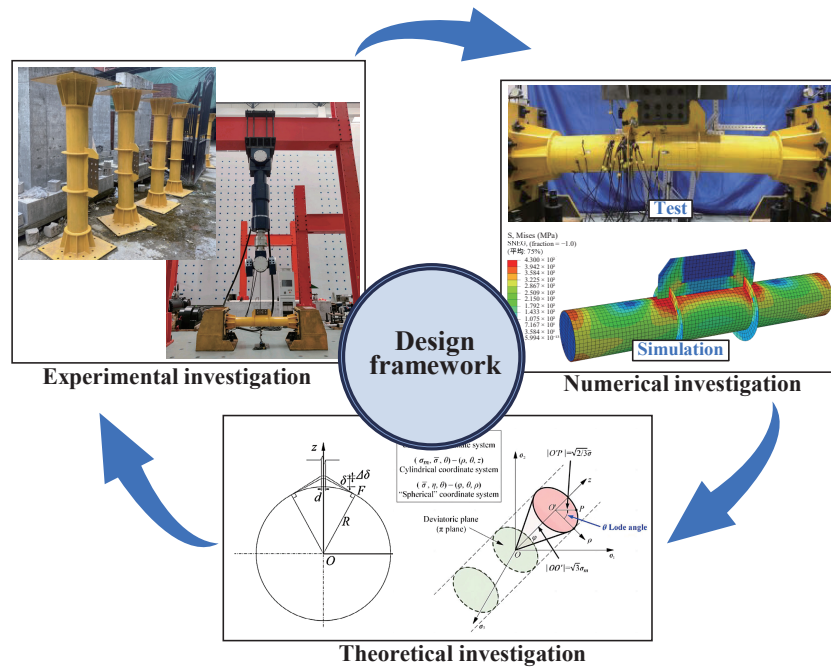


Fig. 16 Schematic diagram of investigations on tube–gusset plate joints.

tests and fatigue tests indicate that in tube–gusset plate joints, the maximum SCF is generally located at the weld toe area of the gusset plate. The incorporation of concrete and ring stiffener plates can significantly reduce the SCF values. Fatigue cracks commonly initiate at points of the maximum hot spot stress, which are typically found in the weld toe area of the gusset plate. In addition, different types of tube–gusset plate joints exhibit distinct crack propagation patterns. The utilization of concrete and ring stiffener plates can substantially improve the fatigue life of joints and slow the rate of crack propagation.

Static tests on tube–gusset plate joints subjected to fatigue damage revealed that the use of concrete and ring stiffener plates effectively enhances the bearing capacity of these joints.

3.4.4 Transition piece

The mechanical performance of SCCT under compression was investigated through experimental, numerical, and theoretical studies, as shown in Fig. 17. A total of 6 SCCT samples and 1 traditional RCT sample were tested [6]. The test parameters included the amount of bottom circumferential rebar, thickness of the stiffened plate, and spacing of the studs. The SCCT samples exhibited a ductile failure mode, as opposed to the brittle failure mode observed in the traditional RCT samples. During the failure of the RCT sample, numerous large-scale fragments and splitting cracks formed in the concrete. In the SCCT samples, deformation at the bottom of the outer and inner steel tubes intensified gradually and was uniformly distributed. Additionally, the interlayer concrete at the corresponding location was crushed, leading to radial displacement of the outer steel tube.

Compared with RCT structures, SCCT structures exhibit significantly greater ultimate load-carrying capacity and ductility. This improvement is attributed primarily to the increased buckling resistance of the steel tubes and the reinforcement provided by the interlayer concrete. Reducing the stud spacing and increasing the thickness of the stiffened plates further enhances the ductility and load-carrying capacity of SCCT structures by strengthening the composite interaction between the concrete and steel. Furthermore, reducing the stud spacing has a pronounced effect on improving the ductility of SCCT.

The parametric analysis identified key factors impacting the ultimate capacity, including the thickness and strength of the interlayer concrete, the thickness of the stiffened plates, the diameter–thickness ratios of the steel tubes, the stud spacing, and the strength of the outer steel tube. The analysis also revealed the relationships between the bearing capacity and these sensitive parameters. A stud spacing of 600 mm was recommended to achieve optimal reinforcement.

The local stability performance of the curved steel plate in SCCT [30] was investigated via a methodology similar to that employed for the tube–gusset plate joint study, as depicted in Fig. 18.

The calculated length coefficients of both the outer and inner steel plates decrease as the ratio of the vertical spacing between the studs and the steel plate thickness (S_v/t) increases. However, this reduction in the calculated length coefficient becomes less pronounced as the S_v/t ratio continues to rise because of the out-of-plane support provided by the concrete. Compared with the outer steel plate, the inner steel plate exhibits better local stability, with a smaller calculation length coefficient for

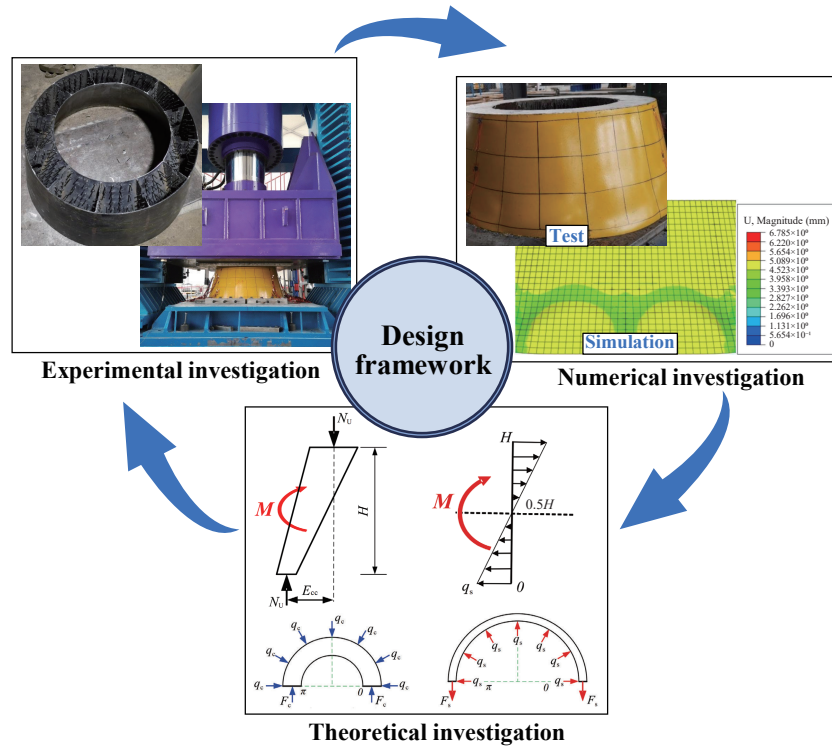


Fig. 17 Schematic diagram of investigations on the mechanical performance of SCCT.

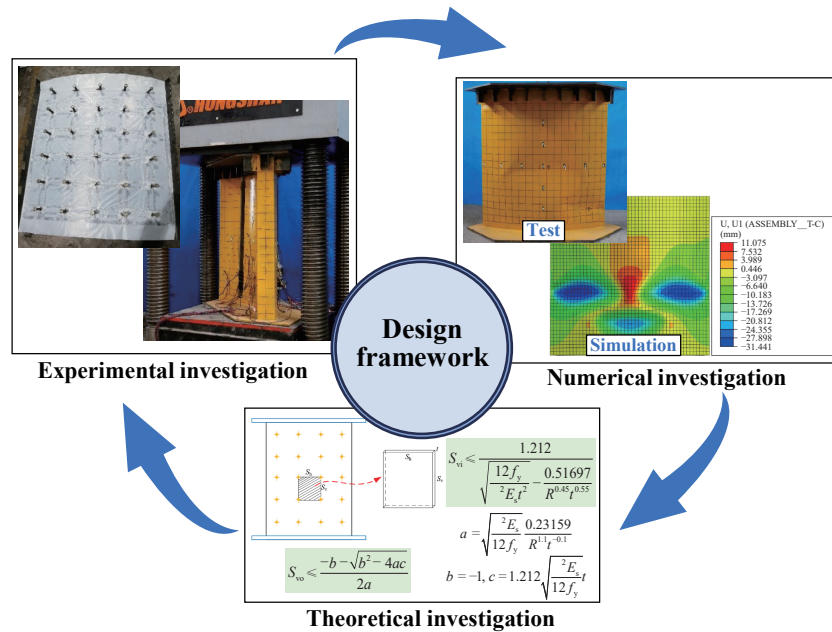


Fig. 18 Schematic diagram of investigations on the local stability performance of curved steel plates in SCCT.

both the inner and outer steel plates. Formulas for calculating the calculation length coefficient of both the inner and outer steel plates are established, along with a proposed method for determining the maximum allowable stud spacing at the curved steel plate–concrete interface.

4 Conclusions

This paper presents innovative studies on steel-pre-

stressed concrete hybrid towers and prestressed CFST lattice towers, which are examined at the component, joint, and structural levels. The main contributions of this study are as follows:

(1) An efficient method for calculating the fundamental frequency of hybrid tall structures was established, revealing the effects of key design parameters on the structural fundamental frequency. In addition, a novel method was also proposed for analyzing wind-induced responses and reverse calculating horizontal thrust from

wind turbines in hybrid high-tower structures.

(2) A comprehensive experimental study on the mechanical performance of hybrid high-towers under complex loading conditions was conducted. This study clarified the force mechanisms under complex stress conditions, established a “layered tube” theoretical model to analyze nonlinear torsional effects, and proposed a method for calculating the load-bearing capacity that accounts for the coupled effects of compression, bending, and torsion.

(3) A construction scheme for SCCTs suitable for hybrid high towers was proposed. On the basis of experimental research and refined numerical analysis, a model was developed to analyze the stability and bearing capacity of arc-shaped steel plates under normal–shear coupled stress. A method was also introduced to determine the spacing limits of studs at the arc-shaped steel plate–concrete interface, revealing the load transfer mechanisms under complex stress conditions in the concrete, inner and outer steel tubes, and through-thickness steel plates.

(4) An experimental study on CFST joints within lattice towers was conducted under both monotonic loading conditions and cyclic loading conditions. This study led to the development of failure criteria and methods for calculating the load-bearing capacity of CFST joints.

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Declaration of competing interest

The authors have no competing interests to declare that are relevant to the content of this article.

Author contribution statement

All authors have given approval to the final version of the manuscript.

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